

Proof of Conjecture 00000001091: the complete 10-arcs of $\text{PG}(2, 11)$ form a single class

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Abstract

Conjecture 00000001091 asserts that the complete 10-arcs of the projective plane $\text{PG}(2, 11)$ form a single isomorphism class. We prove this. Every 10-arc contains a frame, and $\text{PGL}(3, 11)$ is regular on ordered frames, so it suffices to classify the complete 10-arcs through the standard frame $e_1, e_2, e_3, (1, 1, 1)$. The 24 projectivities permuting the frame split the 72 admissible further points into four orbits. A short exhaustive search from one representative of each orbit finds 13 complete 10-arcs, and each is mapped onto a fixed representative by an explicit matrix. The whole argument (linear algebra over $\mathbb{F}_{11}$, the frame lemma, the symmetry reduction, the search with its soundness proof, and the 13 certificates) is formalized in Lean 4 (core library only, no `sorry`, no `native_decide`; axioms `propext` and `Quot.sound` only). An independent Python check finds 84 complete 10-arcs through the frame, all in one orbit, with stabilizer A_5 of order 60. This agrees with the literature: Ball and Lavrauw state that $\text{PG}(2, 11)$ has a unique complete arc of size $q - 1 = 10$.

1 The conjecture and how we read it

Definition: regular completions of arcs. *Conjecture:* the complete 10-arcs of $\text{PG}(2, 11)$ form a single isomorphism class (an entry of the $\text{PG}(2, 11)$ classification, computationally verified).

The Chinese version says the same.

A k -arc of a projective plane is a set of k points, no three of which are collinear. It is *complete* if it is not contained in a $(k + 1)$ -arc, i.e. every point outside the arc lies on a secant (a line joining two points of the arc). The definition line (“regular completions”) only names the topic; the claim to be decided is the conjecture.

Two arcs are *isomorphic* (equivalent) if a collineation of the plane maps one onto the other. The collineation group of $\text{PG}(2, q)$ is $\text{P}\Gamma\text{L}(3, q)$ (fundamental theorem of projective geometry). For $q = 11$ prime, the field $\mathbb{F}_{11}$ has no nontrivial automorphism, so $\text{P}\Gamma\text{L}(3, 11) = \text{PGL}(3, 11)$. Hence “isomorphic” means *projectively equivalent*: some invertible 3×3 matrix over $\mathbb{F}_{11}$ maps one arc onto the other. This is the notion used throughout, and it is the strongest one (a single matrix), so the result also holds for any coarser notion. We prove:

Theorem 1. *Complete 10-arcs exist in $\text{PG}(2, 11)$, and any two of them are projectively equivalent. So the complete 10-arcs form exactly one class.*

2 The model

Points and lines. A point is a 1-dimensional subspace of $\mathbb{F}_{11}^3$, written as its normalized vector (first nonzero coordinate 1). We number the 133 points:

$$11a + b \mapsto (1, a, b) \ (0 \leq a, b \leq 10), \quad 121 + b \mapsto (0, 1, b), \quad 132 \mapsto (0, 0, 1).$$

Lines are numbered the same way by their normalized dual coordinates, and point p lies on line L iff $\text{coord}(p) \cdot \text{coord}(L) = 0$. Three points are *collinear* if they lie on a common line. These 133 dual vectors give *all* lines: any nonzero $\ell \in \mathbb{F}_{11}^3$ is a nonzero scalar multiple of exactly one normalized vector, so by `points_cover` (applied to ℓ) it equals $c \cdot \text{coord}(L)$ for some $L < 133$ and some $c \neq 0$, and has the same zero set. The standard frame is

$$\mathcal{F} = \{e_1, e_2, e_3, u\} = \{0, 121, 132, 12\}, \quad u = (1, 1, 1).$$

Action of matrices. For an invertible matrix M (with inverse N), M maps the point p to the point of $M \text{coord}(p)$, and the line L to the line of $N^T \text{coord}(L)$. The identity

$$(N^T \ell) \cdot (Mx) = \ell \cdot (NMx) = \ell \cdot x$$

shows that incidence is preserved.

Representative. Our representative complete 10-arc is

$$R = \{0, 121, 132, 12, 25, 35, 49, 74, 103, 119\},$$

i.e. the points $(1, 0, 0)$, $(0, 1, 0)$, $(0, 0, 1)$, $(1, 1, 1)$, $(1, 2, 3)$, $(1, 3, 2)$, $(1, 4, 5)$, $(1, 6, 8)$, $(1, 9, 4)$ and $(1, 10, 9)$.

3 Proof of Theorem 1

Lemma 2 (projectivities preserve complete arcs). *Let M be invertible. Then M permutes the points and the lines and preserves incidence. In particular, p, q, r are collinear iff Mp, Mq, Mr are. Consequently, M maps complete k -arcs onto complete k -arcs.*

Proof. The first statements follow from the identity above and $NM = I$. Let A be a complete k -arc. Then MA has k points, since M is injective on points. It is an arc, since collinearity of images implies collinearity of the preimages. For completeness, let $p' \notin MA$ and put $p = Np'$. Then $p \notin A$, so p lies on a secant qr of A , and $p' = Mp$ lies on the secant $(Mq)(Mr)$ of MA . $\square$

Lemma 3 (frame lemma). *Let a_0, a_1, a_2, a_3 be four points, no three collinear. Then some invertible M maps them to e_1, e_2, e_3, u respectively.*

Proof. Classically, write $\text{coord}(a_3) = \sum \lambda_i \text{coord}(a_i)$. All $\lambda_i \neq 0$ by general position. Then the matrix with columns $\lambda_i \text{coord}(a_i)$ maps e_1, e_2, e_3, u to a_0, a_1, a_2, a_3 , and M is its inverse.

In Lean the lemma is applied to the first four entries of the arc's point list. For points of an arc, “no three collinear” is exactly the general-position hypothesis needed here. Lean uses an equivalent four-step construction. Each step is a matrix from an explicit table that was checked exhaustively:

1. $t_1(p)$ maps $p \mapsto e_1$;

2. $t_2(p)$ fixes e_1 and maps $p \mapsto e_2$, for $p \neq e_1$;
3. $t_3(p)$ fixes e_1, e_2 and maps $p \mapsto e_3$, for $p \notin e_1 e_2$;
4. $t_4(p)$ fixes e_1, e_2, e_3 and maps $p \mapsto u$, for p off the triangle.

By Lemma 2, the image of a_k after k steps satisfies the hypothesis of step $k + 1$. $\square$

Every 10-arc contains four points, no three collinear. By Lemmas 2 and 3, every complete 10-arc is therefore equivalent to a complete 10-arc containing $\mathcal{F}$. It remains to classify those.

Candidates and the frame stabilizer. A point of an arc $S \supseteq \mathcal{F}$ outside $\mathcal{F}$ lies on none of the six sides of the quadrangle $\mathcal{F}$. There are exactly 72 such points. The 24 projectivities g_σ mapping (e_1, e_2, e_3, u) to a permutation of itself form a group $G_{\mathcal{F}} \cong S_4$. It acts on the 72 candidates with four orbits:

orbit	rep.	points
O_1 (12)	$25 = (1, 2, 3)$	25, 32, 35, 52, 61, 71, 73, 83, 92, 109, 112, 119
O_2 (24)	$26 = (1, 2, 4)$	26, 28, 29, 31, 39, 43, 46, 51, 63, 65, 68, 69, 75, 76, 79, 81, 93, 98, 101, 105, 113, 115, 116, 118
O_3 (24)	$27 = (1, 2, 5)$	27, 30, 38, 40, 41, 50, 53, 54, 57, 58, 64, 70, 74, 80, 86, 87, 90, 91, 94, 103, 104, 106, 114, 117
O_4 (12)	$37 = (1, 3, 4)$	37, 42, 47, 49, 59, 62, 82, 85, 95, 97, 102, 107

Lemma 4 (symmetry reduction). *Let S be a complete 10-arc containing $\mathcal{F}$. Let j be the least index with $S \cap O_j \neq \emptyset$, and let r_j be the representative of O_j . Then some $g \in G_{\mathcal{F}}$ maps S to a complete 10-arc S' with*

$$S' \supseteq \mathcal{F} \cup \{r_j\} \quad \text{and} \quad S' \cap (O_1 \cup \dots \cup O_{j-1}) = \emptyset.$$

Proof. Such a j exists, since $S \setminus \mathcal{F} \neq \emptyset$ consists of candidates. Pick $c \in S \cap O_j$ and g with $gc = r_j$. Then $g\mathcal{F} = \mathcal{F}$, and the orbits are g -invariant. If $x \in S' \cap O_i$ with $i < j$, then $g^{-1}x \in S \cap O_i$, contradicting the choice of j . Lemma 2 shows that S' is a complete 10-arc. $\square$

Lemma 5 (exhaustive search). *For each $j \in \{1, 2, 3, 4\}$, let S be a complete 10-arc with $S \supseteq \mathcal{F} \cup \{r_j\}$ and $S \cap (O_1 \cup \dots \cup O_{j-1}) = \emptyset$. Then, as a set, S is one of the 13 arcs listed in Table 1.*

Proof. Let $C_j = \mathcal{F} \cup \{r_j\}$. The other five points of S are not in C_j , not in $O_1 \cup \dots \cup O_{j-1}$, and on no secant of C_j . These points form an explicit list rest_j of 43, 37, 22 and 7 points respectively.

The search chooses five of them in list order. After choosing c it discards every later point on a line through c and a chosen point. It also prunes a branch when too few points remain. At a leaf (ten points) it either finds a point on no secant, so the leaf is not complete, or it finds the leaf in Table 1.

The search is sound. At each node:

- the chosen points belong to S ;
- the remaining list contains every point of S that is not chosen;
- the number of points of S in the remaining list equals the number still to be chosen.

j	arc (point numbers; each starts with $\mathcal{F} = 0, 121, 132, 12$)	
1	25, 35, 49, 74, 103, 119	$= R$
1	25, 35, 53, 59, 94, 109	
1	25, 41, 54, 73, 86, 94	
1	25, 41, 70, 95, 109, 119	
1	25, 46, 70, 85, 93, 117	
1	25, 52, 71, 83, 109, 112	
1	25, 53, 65, 68, 85, 104	
2	26, 41, 49, 69, 90, 105	
2	26, 42, 50, 69, 79, 117	
2	26, 42, 82, 98, 101, 113	
2	26, 43, 62, 79, 91, 104	
2	26, 49, 63, 76, 79, 97	
3	27, 57, 74, 86, 94, 106	

Table 1: The 13 complete 10-arcs found by the search of Lemma 5. Each is mapped onto R by an explicit invertible matrix (`orbitData` in the Lean file).

Branching at the first point of S in the list preserves these invariants: a discarded point lies on a line with two chosen points of S , so it is not in S (arc property). At a leaf, S is the chosen set, and it is complete.

The search tree has 816, 497, 138 and 4 nodes, with 28, 6, 22 and 0 leaves. Of these leaves, 7, 5, 1 and 0 are complete. $\square$

Lemma 6 (orbit certificates). *For each arc T of Table 1, an explicit pair of matrices (W, W') with $WW' = W'W = I$ satisfies $WT = R$.*

The matrices are listed in the Lean file and were checked there.

Proof of Theorem 1. R is a complete 10-arc (checked directly). Let A be any complete 10-arc.

1. By Lemmas 3 and 2, some M maps A to a complete 10-arc through $\mathcal{F}$.
2. By Lemmas 4, 5 and 6, a further projectivity maps that arc onto R .

So every complete 10-arc is equivalent to R . As equivalence is symmetric and transitive (inverse and product matrices), any two complete 10-arcs are equivalent. $\square$

4 Further facts and the literature

The Python script `verify.py` computes the following facts. They are not needed for the proof.

- **Arcs through the frame.** For sizes $4, 5, \dots, 12$, the numbers of arcs containing $\mathcal{F}$ are 1, 72, 1548, 10232, 16233, 4032, 336, 72 and 9.
- **Complete ones.** The complete arcs through $\mathcal{F}$ number 40, 5103, 3024, 84, 9 for sizes 7, 8, 9, 10, 12. So the complete arcs of $\text{PG}(2, 11)$ have sizes 7, 8, 9, 10, 12.
- **Incomplete 10-arcs.** There are 84 complete and 252 incomplete 10-arcs through $\mathcal{F}$. Each incomplete one lies on one of the 9 conics through $\mathcal{F}$, and $9\binom{8}{6} = 252$. For example, $C_{10} = \{0, 121, 132, 12, 25, 42, 62, 68, 81, 93\}$ is a 10-arc whose completion to a conic adds the points 109 and 116. The Lean theorem `C10_not_complete` proves that C_{10} is not complete.

- **One orbit.** The 84 complete 10-arcs through $\mathcal{F}$ have the same canonical form (least image over all 5040 frame maps).
- **Stabilizer.** The stabilizer of a complete 10-arc has order 60, with element orders $1^1 2^{15} 3^{20} 5^{24}$, those of A_5 . By orbit counting, its orbit contains $5040/60 = 84$ frame-containing arcs, i.e. all of them.
- **Total count.** The number of complete 10-arcs in $\text{PG}(2, 11)$ is $|\text{PGL}(3, 11)|/60 = 212427600/60 = 3540460$.

This agrees with the literature. Ball and Lavrauw [1] (Corollary 8, based on Hirschfeld’s classification [2, Table 9.4]) state that for $q = 11$ there is a unique complete arc of size $q - 1 = 10$. Their Example 3 is

$$\{(1, 0, 0), (0, 1, 0), (0, 0, 1), (1, 8, 7), (1, 5, 2), (1, 3, 8), (1, 4, 1), (1, 7, 6), (1, 9, 4), (1, 10, 5)\}.$$

`verify.py` checks that this set is a complete 10-arc with the same canonical form as ours. By Segre’s theorem [3] the 12-arcs are the conics, and the enumeration shows that each of the 72 11-arcs through $\mathcal{F}$ lies on one of the 9 conics through $\mathcal{F}$ ($72 = 9 \cdot 8$). So a 10-arc contained in a larger arc is a subset of a conic, consistent with the 252 incomplete ones above.

5 Lean formalization

The project `lean4/` (Lean 4.19.0, core library only) builds in about 85 seconds.

Definitions.

- **Linear algebra.** `F = Fin 11`, vectors `V`, matrices `Mat`, `mulVec`, `mmul`, `dot`; and `IsInv M N` (two-sided inverse).
- **Points and incidence.** `coord` (the 133 points), `inc`, and `Collinear` (a common line).
- **Arcs.** `IsArc`, `IsComplete` and `IsCompleteArc k A`, where A is a list of k distinct points.
- **Equivalence.** `act`, `MapsOnto M A B` (meaning $\{Ma : a \in A\} = B$), and `ProjEquiv`.

Main theorem. `conjecture_00000001091` states

$$(\exists A, \text{IsCompleteArc } 10 \ A) \wedge \forall A \ B, \text{IsCompleteArc } 10 \ A \rightarrow \text{IsCompleteArc } 10 \ B \rightarrow \text{ProjEquiv } A \ B.$$

General lemmas (for all matrices and points, no enumeration).

- **Algebra.** `mulVec_mmul`, `mmul_assoc`, `dot_mulVec` and `tr_mmul`. These are proved from field facts of $\mathbb{F}_{11}$, each checked by `decide`.
- **Action.** `act_comp` and `act_inv`.
- **Collinearity and arcs.** `collinear_act`, `collinear_of_act` and `map_complete`.
- **Search.** The soundness theorem `search_sound` and `search_root`, with the counting lemmas.
- **Assembly.** `reduce`, `to_rep`, `frame_lemma` and `complete10_equiv`.

Kernel-checked finite facts (decide +kernel).

- `pt1_check`: the line masks agree with `inc` on all 133^2 pairs (with `inc_iff`, proved algebraically).
- `T1ok–T4ok`: the frame-lemma tables.
- `g1ok–g4ok`: the frame stabilizer data.
- `cand_cover`, `cover1–cover4` and `rest_ok`.
- `search1–search4`: the four searches.
- `orbit_ok`: the 13 certificates.
- `R_complete`.

Model sanity. The theorems `points_cover`, `points_distinct` and `coord_ne_zero` show that the 133 listed points are exactly the points of $\text{PG}(2, 11)$. `C10_not_complete` shows that completeness is a genuine restriction.

Axioms. `#print axioms` reports only `propext` and `Quot.sound`. There is no `sorry`, no `native_decide` and no added axiom.

Limitations. Collinearity is defined as “lying on a common line” (the incidence definition of $\text{PG}(2, 11)$). The equivalence with the vanishing of the 3×3 determinant is standard, but it is not proved in Lean. The facts of Section 4 (the count 84, the stabilizer A_5 , the size spectrum) are checked only by `verify.py`.

6 Independent check

`verify.py` uses only the Python standard library and no Lean table. It uses a different normalization (last nonzero coordinate 1) and decides collinearity by the determinant. It:

1. enumerates all arcs through the frame;
2. re-checks the 84 complete 10-arcs, and that each of the 252 incomplete ones lies on a conic;
3. computes the canonical forms (1 class) and the stabilizer (order 60, A_5);
4. checks the Ball–Lavrauw example;
5. reproduces the statistics of the Lean search (1455 nodes, 56 leaves, 13 complete).

It runs in about 20 seconds.

7 Reproduction

```
cd lean4 && lake build          # Lean 4.19.0, core only
cd .. && python3 verify.py
pdflatex report.tex && pdflatex report.tex
```

References

- [1] S. Ball and M. Lavrauw, Planar arcs, *J. Combin. Theory Ser. A* (2018); arXiv:1705.10940.
- [2] J. W. P. Hirschfeld, *Projective Geometries over Finite Fields*, 2nd ed., Oxford University Press, 1998.
- [3] B. Segre, Ovals in a finite projective plane, *Canad. J. Math.* 7 (1955), 414–416.